

Problem 1 (20 points). The following are the fourth powers of natural numbers in ascending order:

1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .

The first ten entries of this sequence are spelt out in Ik, in arbitrary order:

- a. **ɲamɪ́rá lebetse ńda tomínékwa túde ńda jírɪ́nɪ túde ńda kíɖɪ kɔ́n**
- b. **álifika ts'agúsé ńda tomínékwa túde ńda kíɖɪ ts'agúsé ńda jírɪ́nɪ túde ńda kíɖɪ kɔ́n**
- c. **álifika túde ńda kíɖɪ kɔ́n ńda ɲamɪ́rá túde ńda tomínékwa túde ńda kíɖɪ kɔ́n ńda nébɛɛ kɔ́n**
- d. **álifika tomín**
- e. **álifa kɔ́n ńda ɲamɪ́rá lebetse ńda tomínékwa túde ńda kíɖɪ ts'agúsé ńda jírɪ́nɪ túde ńda kíɖɪ kɔ́n**
- f. **tomínékwa túde ńda kíɖɪ aɖe ńda nébɛɛ kɔ́n**
- g. **tomíní ńda jírɪ́nɪ túde ńda kíɖɪ kɔ́n**
- h. **kɔ́n**
- i. **álifika lebetse ńda ɲamɪ́rá ts'agúsé ńda kɔ́n**
- j. **ɲamɪ́rá túde ńda kíɖɪ kɔ́n ńda tomínékwa lebetse ńda jírɪ́nɪ túd**

(a) Determine the correct correspondence.

(b) Write in numerals:

- k. **álifika tomíní ńda jírɪ́nɪ lebetse**
- l. **tomínékwa túde ńda kíɖɪ kɔ́n**
- m. **tomínékwa túde ńda nébɛɛ kɔ́n**
- n. **álifika ɲamɪ́rá kɔ́n**

(c) Write out in Ik: 3108.

(d) Write out the eleventh entry of the sequence in Ik.

△ Ik belongs to the Nilo-Saharan family. It is spoken by approx. 7,500 people in Uganda.

ɓ, **ɗ**, **j**, **ɲ** and **ts'** are consonants. **ɔ**, **ɛ** and **ɪ** are pronounced like **o**, **e** and **i**, respectively, but with the slightly lowered tongue. The mark ´ denotes a high tone.

The fourth power of a number n is the result of multiplying four instances of n together.
($n^4 = n \times n \times n \times n$) —Minkyu Kim